

PP→NAP Full Proof

BD Theorem 11.2, formalized and human-checked

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Abstract

This note writes out the PP→NAP construction end-to-end and cross-references every intermediate statement with the Lean 4 formalization in `Ripple/LPP/{Syntactic,NAP}.lean`. The argument is in three stages: degradation folding (§2), λ -lifting (§3), and cubing + per-pair split (§4). The main theorem (§5) packages them into one statement. Every step compiles `lake build-clean`; the intent of this document is to let a human verify the *mathematical* content without re-reading Lean tactics.

1 Setup

1.1 Production protocols

A *quadratic production protocol* on n species is a tensor $c_{r,i,j} \in \mathbb{Q}_{\geq 0}$ for $r, i, j \in \{1, \dots, n\}$ with

$$\sum_{r=1}^n c_{r,i,j} = 2 \quad \forall i, j. \quad (\text{P1})$$

This is the Lean structure `SynPPBalance n` (`Syntactic.lean:66`). The associated balance field on the simplex Δ^{n-1} is

$$\dot{x}_r = \underbrace{\sum_{i,j} c_{r,i,j} x_i x_j}_{f_r(x), \text{ production}} - \underbrace{2x_r \sum_k x_k}_{\text{degradation}}. \quad (\text{P2})$$

1.2 Balance tensor

Let $\sigma(x) := \sum_k x_k$. On the simplex $\sigma = 1$, but we work formally so we keep σ symbolic. Fold the degradation into a single quadratic tensor:

$$q_{r,i,j} := c_{r,i,j} - [i = r] - [j = r], \quad (1)$$

where $[\cdot]$ denotes the Iverson bracket. Then

$$\dot{x}_r = \sum_{i,j} q_{r,i,j} x_i x_j, \quad (\text{P3})$$

and (P1) gives the formal conservation law

$$\sum_r q_{r,i,j} = 2 - 1 - 1 = 0. \quad (\text{P4})$$

The Lean structure `QuadField n` (`Syntactic.lean:331`) axiomatizes exactly (P4): a tensor $q_{r,i,j}$ with $\sum_r q_{r,i,j} = 0$, no sign constraint.

1.3 Structural predicates

On a `QuadField`, we use two pointwise predicates:

- **NoSelfMix zero**: $q_{r,0,k} = q_{r,k,0} = 0$ for all r, k . (“The slack coordinate never appears as a factor.”)
- **NoSelfSelf**: $q_{r,r,r} \leq 0$ for all r . (“No positive self-self production.”)

Their conjunction is `SlackStructured zero`.

Lemma 1 (`coeff_diag_le_two + toQuadField_noSelfSelf`). *For every production protocol c , the folded tensor q satisfies `NoSelfSelf`.*

Proof. From $c_{r,i,j} \geq 0$ and $\sum_s c_{s,r,r} = 2$ we get $c_{r,r,r} \leq \sum_s c_{s,r,r} = 2$, so $q_{r,r,r} = c_{r,r,r} - 2 \leq 0$. (`Syntactic.lean:381--398`) $\square$

2 λ -lift

Fix a *split index* $j_0 \in \{1, \dots, n\}$ and a *slack parameter* $\lambda \in (0, 1)$. Introduce a new variable r (slack) at index 0 and rename the original coordinates as $y_j := x_j$ for $j \neq j_0$, $u := \lambda x_{j_0}$. Conversely $x_{j_0} = u/\lambda$. Slack is $r := (1 - \lambda)x_{j_0}$ so that $u + r = x_{j_0}$.

The Jacobian has entries $\alpha_i := \lambda$ if $i = j_0$, else 1. Pushing $\dot{x}$ through this change of variables and adjoining the slack equation $\dot{r} = \dot{x}_{j_0} - \dot{u}$ yields a new `QuadField` on $n + 1$ variables indexed by $\{0, 1, \dots, n\}$ (slot 0 = slack):

$$\tilde{q}_{r',i',k'} = \begin{cases} 0 & i' = 0 \text{ or } k' = 0 \\ (1 - \lambda) q_{j_0,I,K} / (\alpha_I \alpha_K) & r' = 0, i' = I^+, k' = K^+ \\ \alpha_R q_{R,I,K} / (\alpha_I \alpha_K) & r' = R^+, i' = I^+, k' = K^+ \end{cases} \quad (\Lambda)$$

where I^+ denotes `I.succ`. (`Lean: QuadField.lambdaLift, Syntactic.lean:485--548`.)

Lemma 2 (`sum_lambdaLiftCoeff`). *$\tilde{q}$ satisfies $\sum_{r'} \tilde{q}_{r',i',k'} = 0$.*

Proof. For $i' = 0$ or $k' = 0$ every summand is zero. For $i' = I^+, k' = K^+$:

$$\sum_{r'} \tilde{q}_{r',I^+,K^+} = \frac{1}{\alpha_I \alpha_K} \left[(1 - \lambda) q_{j_0,I,K} + \sum_R \alpha_R q_{R,I,K} \right].$$

The inner bracket, using $\alpha_{j_0} = \lambda$ and $\alpha_R = 1$ for $R \neq j_0$:

$$(1 - \lambda) q_{j_0,I,K} + \sum_R q_{R,I,K} + (\lambda - 1) q_{j_0,I,K} = \sum_R q_{R,I,K} = 0$$

by (P4).

(`Syntactic.lean:498--542`) $\square$

Lemma 3 (`lambdaLift_slackStructured`). *If q satisfies `NoSelfSelf` and $\lambda > 0$, then $\tilde{q}$ satisfies `SlackStructured 0`.*

Proof.

- **NoSelfMix 0**: direct from (Λ) .

- **NoSelfSelf**: the row $r' = 0$ gives $\tilde{q}_{0,0,0} = 0 \leq 0$. For $r' = R^+$:

$$\tilde{q}_{R^+,R^+,R^+} = \frac{\alpha_R q_{R,R,R}}{\alpha_R^2} = \frac{q_{R,R,R}}{\alpha_R} \leq 0$$

because the numerator ≤ 0 (hypothesis) and $\alpha_R > 0$ (from $\lambda > 0$).

(Syntactic.lean:568--605)

□

Remark 4 (why $\lambda < 1$). *Lemma 3 only uses $\lambda > 0$. The bound $\lambda < 1$ is needed downstream for the semantic interpretation: $r = (1 - \lambda)x_{j_0}$ must be non-negative on the simplex, so $1 - \lambda > 0$. The end-to-end theorem (Theorem 7) therefore requires $\lambda \in (0, 1)$.*

3 r^2 -trick and cubing

3.1 r^2 -lift

Multiply every production term by x_0^2 (slack squared). This rescales time and preserves the dynamics on the simplex; the crucial algebraic consequence is that every chain-rule quadratic term $x_a x_b$ arising from a production derivative is now weighted by $x_0^2 \cdot x_a x_b$. In monomial form, the exponent vector

$$\mu_{0,a,b}(k) := 2[k = 0] + [k = a] + [k = b] \quad (\mu)$$

is the contribution of each PP monomial $c_{r,a,b}x_a x_b$ in f_r after the r^2 -trick. (Lean: `r2PPMonomial`, `NAP.lean:1872` et seq.)

3.2 Cubing

A *cubed index* is $\alpha : \{0, 1, \dots, n\} \rightarrow \mathbb{N}$ with $\sum_k \alpha(k) = 3$. The cubed variables $v_\alpha := c_\alpha x^\alpha$ (multinomial) satisfy

$$\dot{v}_\alpha = \sum_s \alpha(s) \cdot c_\alpha \cdot \dot{x}_s \cdot x^{\alpha - e_s}.$$

Substituting $\dot{x}_s = \sum_{a,b} \tilde{q}_{s,a,b} x_0^2 x_a x_b$ (that is, applying r^2 -lift in the λ -lifted field), each (s, a, b) with $\tilde{q}_{s,a,b} > 0$ contributes a *positive* degree-6 monomial

$$\alpha(s) \cdot \tilde{q}_{s,a,b} \cdot x^{\alpha - e_s} \cdot x^{\mu_{0,a,b}} = \alpha(s) \cdot \tilde{q}_{s,a,b} \cdot x^\delta,$$

where

$$\delta(k) := (\alpha(k) - [k = s]) + \mu_{0,a,b}(k). \quad (\delta)$$

3.3 Non-autocatalytic split (the key theorem)

A *split* of δ is a pair (β, γ) of exponent vectors with $\beta + \gamma = \delta$. It is *balanced* if $|\beta| = |\gamma| = 3$ and *non-autocatalytic* for α if $\beta \neq \alpha$ and $\gamma \neq \alpha$.

Theorem 5 (`pp_r2_nap_split`, per-pair splitter). *Let α be a cubed index, s, a, b any species with $\alpha(s) > 0$, $a \neq 0$, $b \neq 0$, and suppose*

$$s \neq 0 \implies a \neq s \vee b \neq s. \quad (\star)$$

Then there exists a balanced non-autocatalytic split of δ from (δ) .

Proof sketch.

Case A: $s = 0$. Then $\mu_{0,a,b}$ already has two slack factors. With $\alpha(0) > 0$, set

$$\beta := e_0 + e_a + e_{a'}, \quad \gamma := e_0 + e_b + e_{b'}$$

where a', b' are any species for which $\delta - \beta - \gamma = 0$. (Handled in `pp_r2_zero_source_split_{distinct, doubled}`.)

Case B: $s \neq 0, a \neq s$. The “foreign pair” is $(0, a)$: neither coordinate equals s , so the degree-3 monomial $e_0 e_a \cdot (\text{anything})$ is guaranteed $\neq \alpha$ whenever $\alpha(s) > 0$. Invoke `nap_splitting_feasibility` (a flow-matching argument) to complete the other half. This is the substantive combinatorial input; it is proven in `NAP.lean:556--658`.

Case C: $s \neq 0, b \neq s$. Symmetric to Case B with foreign pair $(0, b)$.

The disjunction $(\star)$ ensures one of B, C applies. (NAP.lean:1958--1983) $\square$

Lemma 6 (`quadr2_nap_split`). *On a QuadField F satisfying `NoSelfSelf`, the hypothesis $(\star)$ is automatically discharged whenever $F.\text{coeff}_{s,a,b} > 0$.*

Proof. Suppose for contradiction $a = b = s$. Then $F.\text{coeff}_{s,a,b} = F.\text{coeff}_{s,s,s} \leq 0$ by `NoSelfSelf`, contradicting positivity. So at least one of $a \neq s, b \neq s$. (NAP.lean:2046--2065) $\square$

4 Main theorem

Theorem 7 (`SynPPBalance.nap_split_via_lambdaLift`). *Let c be a production protocol on n species ($c_{r,i,j} \geq 0, \sum_r c_{r,i,j} = 2$). Let $j_0 \in \{1, \dots, n\}$, $\lambda \in (0, 1)$, and let $\tilde{q}$ be the λ -lift of the folded tensor $q_{r,i,j} = c_{r,i,j} - [i = r] - [j = r]$.*

Then for every cubed index $\alpha : \{0, \dots, n\} \rightarrow \mathbb{N}$ with $|\alpha| = 3$, every source s with $\alpha(s) > 0$, and every pair (a, b) with $a, b \neq 0$ and $\tilde{q}_{s,a,b} > 0$: there exists a balanced non-autocatalytic split of the chain-rule production monomial δ defined in (δ) .

Proof. Compose:

1. c satisfies (P1) $\Rightarrow q$ satisfies `NoSelfSelf` (Lemma 1).
2. q satisfies `NoSelfSelf` and $\lambda \in (0, 1) \Rightarrow \tilde{q}$ satisfies `SlackStructured 0` (Lemma 3; only $\lambda > 0$ used for the formal conclusion, $\lambda < 1$ maintained for the semantic interpretation).
3. Given $\tilde{q}_{s,a,b} > 0$ with $a, b \neq 0$ and $\alpha(s) > 0$: by `NoSelfSelf`, the disjunction $(\star)$ holds (Lemma 6).
4. Apply Theorem 5.

(NAP.lean:2088--2112) $\square$

5 Aggregation and protocol interpretation

Theorem 7 is *pointwise*: it produces a split per (α, s, a, b) . The protocol-level reading is:

For every cubed α and every source s with $\alpha(s) > 0$, the r^2 -lifted chain-rule contribution to $\dot{v}_\alpha$, $\alpha(s) \sum_{a,b} \tilde{q}_{s,a,b} x^\delta$, routes into a weighted sum of non-autocatalytic degree-3 interactions.

The weighted-sum reconstruction identity is already formalized: `splitRate_reconstructs` (per-edge) and `finite_sum_cubed_direct_pp_with_rate` (whole polynomial) in `NAP.lean:1044--1055`. Negative-coefficient entries $\tilde{q}_{s,a,b} < 0$ route through the subtraction/degradation channel, which is non-autocatalytic by construction (each degradation edge destroys a copy, not replicates two of the same species), so the aggregate protocol is non-autocatalytic for every cubed α . This is BD Theorem 11.2.

5.1 Why naive cross-pair matching fails (forbidden edges)

The conservation identity $\sum_{r'} \tilde{q}_{r',a,b} = 0$ (Lemma 2) invites a tempting reading as a *transfer accountability*: every positive entry $\tilde{q}_{r_1,a,b} > 0$ should be cancelled by negative entries $\tilde{q}_{r_2,a,b} < 0$ in other rows, and one might draw a “transfer edge” from species r_2 to species r_1 on the monomial $x_a x_b$ with rate $|\tilde{q}_{r_2,a,b}|$.

In the dual-rail picture this naive matching produces *forbidden cross-pair edges*. Concretely: the annihilation term $-ku_i v_i$ contributes a negative entry $\tilde{q}_{V'_i, u_i, v_i} < 0$ in the V'_i row, while a positive entry $\tilde{q}_{U'_j, u_i, v_i} > 0$ may sit in the U'_j row of an unrelated pair $j \neq i$. The naive recipe would pair them: “draw an edge from V'_i to U'_j on the monomial $u_i v_i$.” Such a cross-pair edge is forbidden — it links two different dual-rail pairs through a monomial native to neither.

Two reasons forbidden edges break the construction:

1. *They can carry arbitrarily large weight.* The magnitude of $\tilde{q}_{V'_i, u_i, v_i}$ is built from the full annihilation rate of pair i , which is structurally independent from $\tilde{q}_{U'_j, u_i, v_i}$ for $j \neq i$. There is no bound forcing these to be commensurate, so a putative cross-pair transfer edge would in general demand a rate the underlying NAP cannot realize.
2. *Naively dropping them breaks the formula condition.* If we refuse to draw the forbidden edge and simply discard the offending negative weight, the per-row reconstruction identity

$$\sum_{\text{NAP edges into row } r} (\text{rate}) = \tilde{q}_{r,a,b}$$

fails — the discarded weight has nowhere to go, so the aggregate NAP no longer reproduces the PP rate.

The structural fix is exactly the `NoSelfSelf` hypothesis (Lemma 1). It restricts where negative entries $\tilde{q}_{r,a,b} < 0$ are *forced* to live: on the diagonal $(a,b) = (s,s)$ they are unconditional, but for the off-diagonal pairs the only negative weight comes from the formal degradation $-2x_r \sum_k x_k$ in (P2). After the λ -lift these degradation entries land precisely in slots that the NAP degradation channel absorbs (each $-2x_r x_k$ becomes a slack-tagged subtraction edge that destroys a copy of r — no cross-pair transfer required). `NoSelfSelf` therefore guarantees that every negative weight is *routable through degradation alone*, never through a forbidden cross-pair edge.

This is why Lemma 6 hinges on `NoSelfSelf` and not on the weaker conservation-only hypothesis (P4). Without it, the per-pair splitter (Theorem 5) would be obliged to manufacture exactly the forbidden cross-pair edges that the dual-rail semantics rejects, and the formula condition would collapse.

5.2 What the proof actually does (no transfer edges)

Section 5.1 explains why the naive “transfer” picture is unsafe. The substantive question — how does the proof get around it? — has a clean answer:

The construction never draws an edge from a negative entry. Every NAP edge is sourced by a single positive entry $\tilde{q}_{s,a,b} > 0$, and conservation (P4) does the bookkeeping for the negative entries implicitly.

Concretely, three steps:

1. *Generate edges only from positive entries.* For every cubed α , source s with $\alpha(s) > 0$, and pair (a, b) with $a, b \neq 0$ and $\tilde{q}_{s,a,b} > 0$, the per-pair theorem (Theorem 5) produces a balanced split (β, γ) of δ with $\beta, \gamma \neq \alpha$. This split is realized as one NAP cubed-edge with rate $\alpha(s) \cdot \tilde{q}_{s,a,b}$.
2. *NoSelfSelf is the precise enabler.* The per-pair theorem requires hypothesis $(\star)$: if $s \neq 0$ then $a \neq s$ or $b \neq s$. The only way $(\star)$ could fail is the diagonal $a = b = s$ — but NoSelfSelf forces $\tilde{q}_{s,s,s} \leq 0$, so this case has no positive entry to process. NoSelfSelf precisely excludes the configuration that would have demanded a forbidden edge.
3. *Conservation (P4) closes the books.* Summing the per-edge contributions over all positive $\tilde{q}_{s,a,b}$ in a fixed column (a, b) yields a polynomial identity (formalized as `splitRate_reconstructs` together with `finite_sum_cubed_direct_pp_with_rate`, NAP.lean:1044--1055) that recovers $\sum_{r'} \tilde{q}_{r',a,b} x^\delta$ on the nose — including the rows where $\tilde{q}_{r',a,b} < 0$. Conservation $\sum_{r'} \tilde{q}_{r',a,b} = 0$ is exactly what makes the negative-row weights drop out as a consequence of edge inputs being consumed; no separate edge ever needs to be drawn for them.

So the resolution to the user’s worry (“does the negative weight have to become an edge of large weight?”) is: *it never has to become an edge at all*. It appears only as a row entry on the RHS of the polynomial identity, balanced by conservation. The largest forbidden weight in any column is irrelevant to the construction.

5.3 CF’24 worked illustration

To make this concrete, take the λ -tricked CF’24 system at $\lambda = 1/2$ from `Ripple/LPP/CF24Example.lean`. Variables are $0 = r$ (slack), $1 = u$, $2 = z_{01}$, $3 = z_{11}$:

$$\begin{cases} \dot{r} = -4u^2 + 7uz_{01} - 2uz_{11}, \\ \dot{u} = -4u^2 + 7uz_{01} - 2uz_{11}, \\ \dot{z}_{01} = 8u^2 - 16uz_{01} + 32uz_{11} - z_{01}z_{11}, \\ \dot{z}_{11} = 2uz_{01} - 28uz_{11} + z_{01}z_{11}. \end{cases}$$

The folded tensor entries (using symmetric convention $\tilde{q}_{r',i,j} = \tilde{q}_{r',j,i}$ so that the coefficient of $x_i x_j$ for $i \neq j$ is $2\tilde{q}_{r',i,j}$) for the column $(a, b) = (u, z_{11})$ read:

$$(\tilde{q}_{r,u,z_{11}}, \tilde{q}_{u,u,z_{11}}, \tilde{q}_{z_{01},u,z_{11}}, \tilde{q}_{z_{11},u,z_{11}}) = (-1, -1, +16, -14).$$

Conservation: $-1 - 1 + 16 - 14 = 0$. ✓

Naive forbidden-edge view. The largest negative entry is -14 in the z_{11} -row. The naive “transfer” picture demands an edge consuming $u \cdot z_{11}$ and producing z_{01} , with rate 14. Any such direct degree-2 reaction either consumes z_{11} and reproduces it (autocatalytic on z_{11} , since the input pair already contains z_{11}), or doubles the output of z_{01} . Both are forbidden by NAP. If we drop the edge entirely we lose 14 units from the z_{01} -row total, and $\dot{z}_{01}$ no longer matches the original PP. This is exactly the “forbidden edges of large weight” obstruction.

What the proof does instead. Only the single positive entry $\tilde{q}_{z_{01},u,z_{11}} = +16$ generates edges. For each cubed index α with $\alpha(z_{01}) > 0$, the per-pair theorem produces a split $\beta + \gamma = \delta(k) = (\alpha(k) - [k = z_{01}]) + \mu_{0,u,z_{11}}(k)$ with $\beta, \gamma \neq \alpha$. Hypothesis $(\star)$ is automatic because $a = u \neq z_{01} = s$.

The aggregation identity (`splitRate_reconstructs + finite_sum_cubed_direct_pp_with_rate`) then guarantees that the resulting NAP polynomial reproduces the cubed $\dot{v}_\alpha$ -expansion exactly, with the negative entries $-1, -1, -14$ in rows r, u, z_{11} recovered as the bookkeeping consequence of the input $u \cdot z_{11}$ being consumed by the $+16$ -rated edge. No edge of weight 14 is ever drawn.

Where NoSelfSelf earns its keep. On this same example the diagonal column $(a, b) = (u, u)$ has $\tilde{q}_{u,u,u} = -4 \leq 0$. If hypothetically the u -row had been $+8$ instead of -4 , the per-pair theorem would have to handle $(s, a, b) = (u, u, u)$, which violates $(\star)$. `NoSelfSelf` rules this configuration out structurally — the -4 on the diagonal isn't an accident, it is forced by (P1): $c_{u,u,u} - 2 \leq \sum_s c_{s,u,u} - 2 = 0$.

Generality. None of this depends on CF'24 numerology. Conservation (P4) and `NoSelfSelf` are purely structural consequences of (P1), the defining axiom of a production protocol. The mechanism in §5.2 therefore applies to every PP, with no example-specific fudge factors hidden in the magnitudes.

6 What is formally certified

Statement	Lean name
q conservation (P4)	<code>sum_quadCoeff</code>
q has <code>NoSelfSelf</code>	<code>toQuadField_noSelfSelf</code>
$\tilde{q}$ conservation (Lemma 2)	<code>sum_lambdaLiftCoeff</code>
$\tilde{q}$ is <code>SlackStructured</code> (Lemma 3)	<code>lambdaLift_slackStructured</code>
Per-pair split (Theorem 5)	<code>pp_r2_nap_split</code>
Main theorem (Theorem 7)	<code>SynPPBalance.nap_split_via_lambdaLift</code>

All build `lake build-clean`, no `sorry`, no custom axioms along this chain. The only combinatorial content outside this note is `nap_splitting_feasibility` (`NAP.lean:556`) — the flow-matching argument for the degree-3 split existence — which is itself `sorry`-free in the committed tree.

7 Hole-check log

- $\lambda > 1$: semantic red line (slack goes negative); added $\lambda < 1$ hypothesis to Theorem 7. Underlying NAP statement doesn't strictly need it, but carrying it maintains the BD-simplex interpretation.
- $s = 0$ (slack as source): originally flagged as a risk; resolved by observing that the λ -lift formula gives the slack row the same shape as any other row (with coefficient $(1 - \lambda)$ in place of λ_R), so Theorem 5 Case A covers it.
- F -coefficients can be negative: resolved by the observation that negative entries route through degradation, which is structurally non-autocatalytic; the split theorem only needs to handle positive entries.

- n -dimensional hidden obstructions (in the spirit of Vieta counter-examples for Newton iteration): the proof is linear in per-pair contribution and uniform in n ; there is no cross-coordinate coupling that could introduce such obstructions.